

Demand: Demand refers to the willingness and ability of buyers and purchase different quantities of a good at different prices during a specific time period.

Wants: Wants are unlimited desire or wishes that people have for good and services.

Quantity Demanded: Quantity demanded is the number of unit of a good that individuals are willing and able to buy at a particular price during a time period.

Law of Demand: (In word) "All other things constant" or "nothing else changes" or *ceteris paribus*, as the price of a good rises, the quantity demanded of the good falls, and as the price of the good fall, quantity demanded will rises.

Demand schedule: A demand table schedule is a table showing how much of a given product a household would be willing to buy at different prices, other thing held constant.

Table: Demand schedule for cornflakes:

Price (Per box)	Quantity Demanded (million of boxes per year)
5	9
4	10
3	12
2	15
1	20

A change in quantity demanded versus a change in demand.

A change in quantity demanded: A movement from one point to another point on the same demand curve caused by a change in the price of the good.

Demand schedule - (A)

Price (\$)	Q _d (Units)
\$ 20	500
\$ 15	600
\$ 10	700
\$ 5	800

Figure : 1: A change in quantity demanded of good x

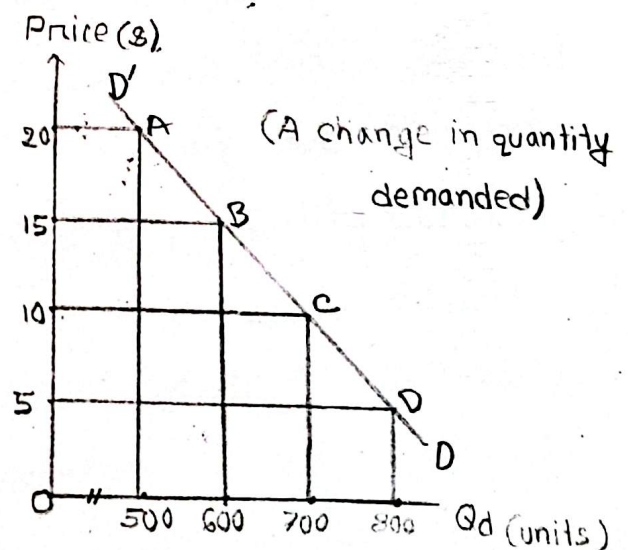


Figure 1, shows us a change in quantity demanded. The horizontal axis labeled 'Quantity demanded' for the good x and the vertical axis labeled 'Price as dollars'. When the price of good x is \$20, the quantity demanded is 500 units. Point A represents it.

Figure: A change in Demand (Increase/decrease)

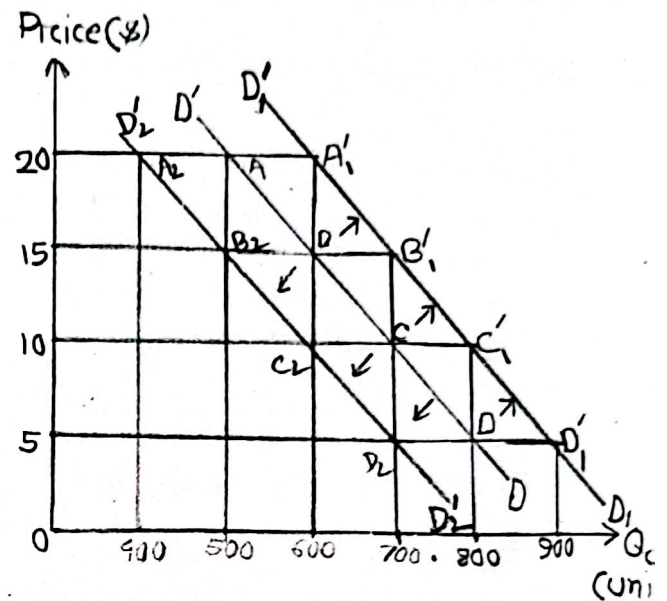
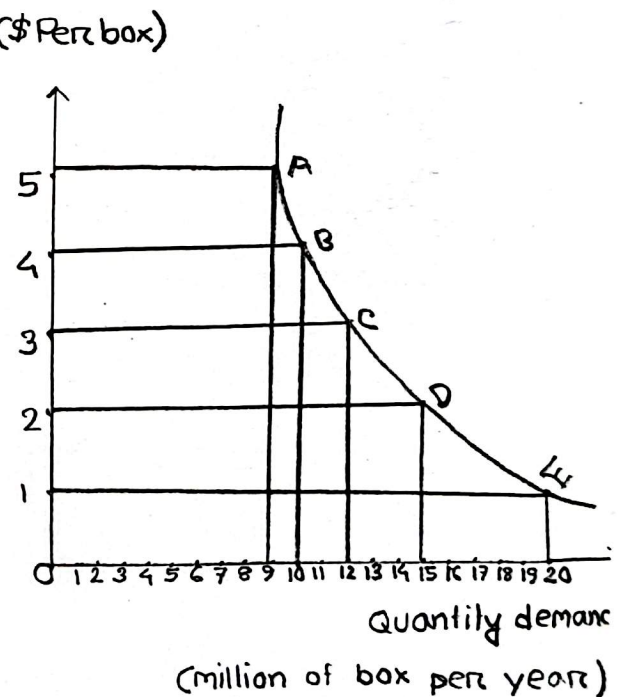


Figure shows a change in demand (increase and decrease). Horizontal axis represents the quantity demanded in unit and vertical axis represents the price in dollar. DD' is the initial demand curve. After rightward shifting the demand curve is D_1D_1' . In that situation, quantity demanded is more than before at the same price level. Such as, at the same price of 20 dollar in point A quantity demanded is 500 unit and in point A' quantity demanded is 600 unit. Thus after leftward ~~sh~~ shifting the demand curve is D_2D_2' .

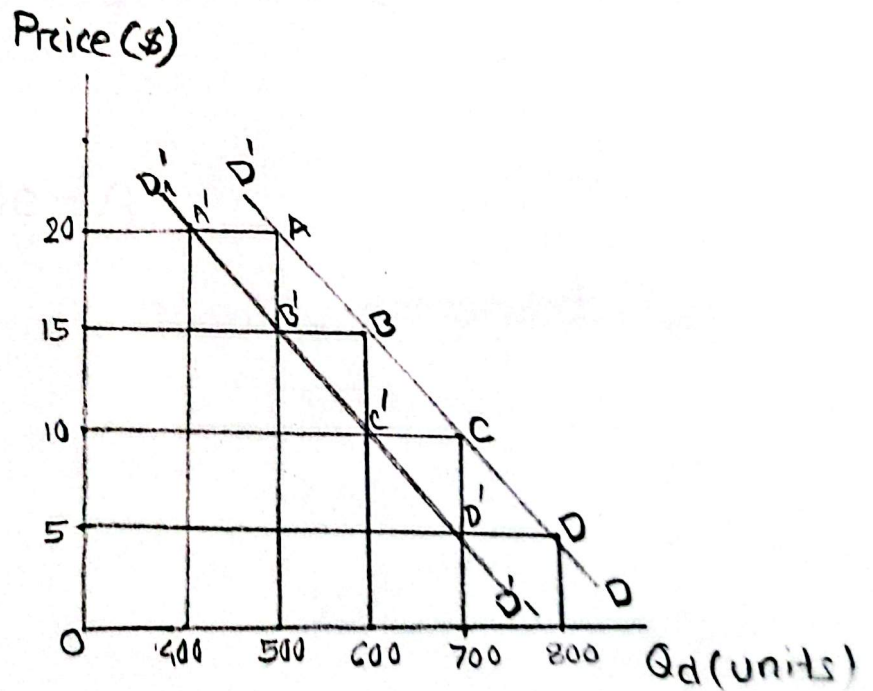
Demand curve: The demand curve is a graph illustrating how much of a given product a household would be willing to buy at different prices, other things held constant.

Figure: 1.1. Showing demand curve.



Demand schedule c (Decrease in Demand)

Price (\$)	Quantity Demanded (units)
20	500 400
15	600 500
10	700 600
5	800 700



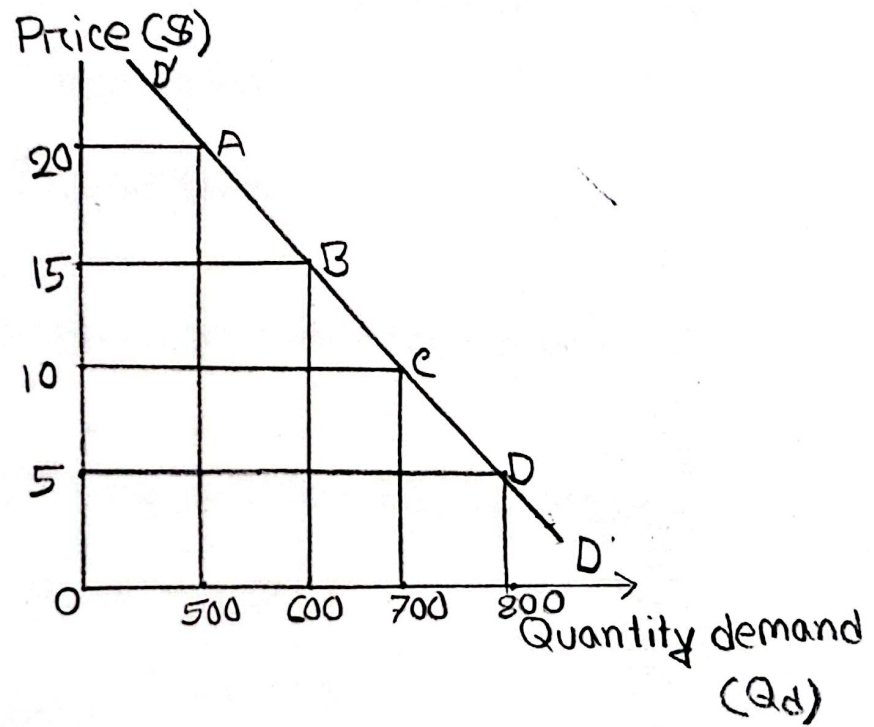
A change in quantity demanded versus a change in demand:

Quantity demanded: The number of units of a good that individuals are willing and able to buy at a particular price during a time period.

A change in quantity demanded: A movement from one point to another point on the same demand curve caused by a change in the price of the good.

Demand schedule - A

Price (\$)	Quantity demanded (Q_d)
20	500
15	600
10	700
5	800



A change in demand / shift in demand curve.

Demand $\rightarrow$ Increase

Decrease

Demand schedule B (Increase in demand)

Price (\$)

Quantity demanded (Units)

20

~~500~~ $\rightarrow$ 600

15

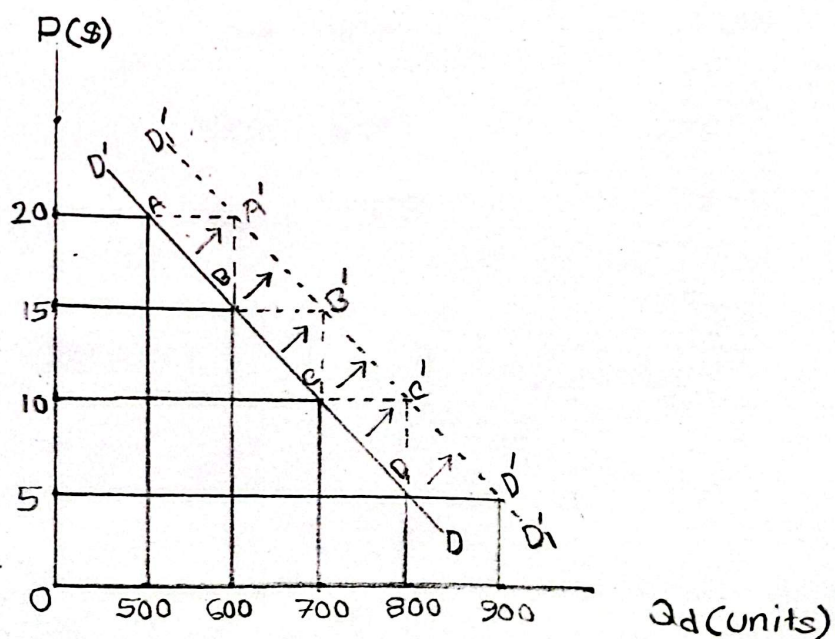
~~600~~ $\rightarrow$ 700

10

~~700~~ $\rightarrow$ 800

5

~~800~~ $\rightarrow$ 900



when the price decrease from 200 dollar to 15 dollar the quantity demanded of good x ~~to~~ increases from 500 unit to 600 unit. Points 'B' indicates it. After that, when the price is 10 dollar, the quantity demanded is 700 unit. At last, we can see that, when the price is 5 dollar the quantity demanded is 800 unit. Points 'D' represents it. All combination are follow the law of demand.

A change in demand / shift in Demand curve:

Demand schedule B (Increase in demand)

price (\$)	(Increased) Qd (units)	(Decreased) Qd (units)
20	500 600	400
15	600 700	500
10	700 800	600
5	800 900	700

A change in Demand:

Law of Downward sloping Demand:

Quantity Demanded tends to fall as price rises for two reasons.

1. Substitution effect: When the price of a good rises, we will substitute other similar good for it.
2. Income effect: When a price goes up we find ourselves somewhat poorer than before.

Individual Demand Curve: An individual demand curve represents the price quantity combinations of a good for a single buyer.

Market Demand Curve: A market demand curve represents the price quantity combinations of a good for all buyers.

A market demand curve is derived by "adding up individual demand curves."

Quantity Demanded				
Price(\$)	John	Smith	Other Buyers	All Buyers
15	1	2	20	23
14	2	3	45	50
13	3	4	70	77
12	4	5	100	109
11	5	6	130	141
10	6	7	160	173

* Demand curve for:-

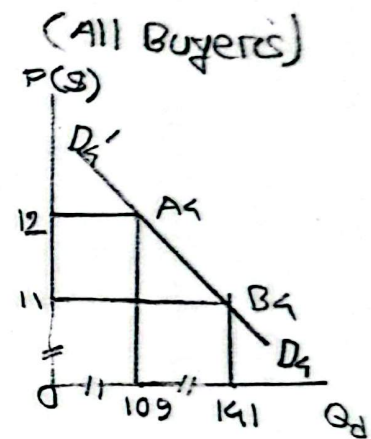
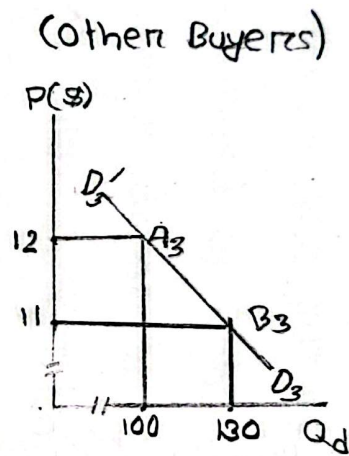
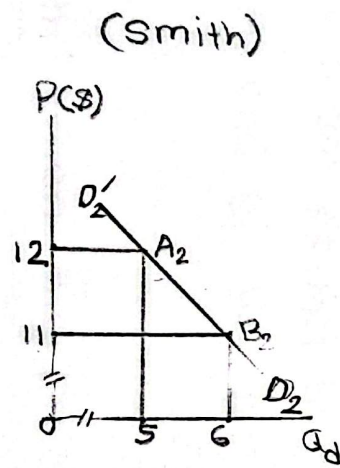
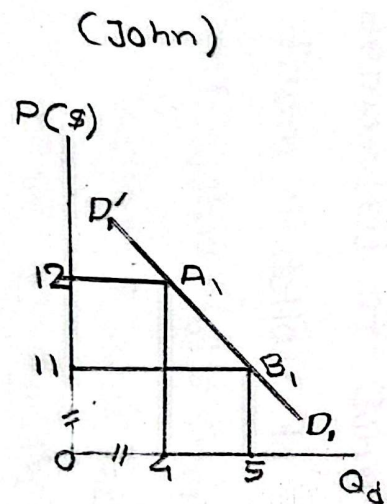


Figure 1.2 shows the horizontal axis as quantity demanded (Q_d) and vertical axis shows as price ($\$$). Here, we can take two different prices for this quantity demanded.

Demand curve for John: when the price is 11 dollar the quantity demanded of John is 5 unit. If prices rises from 11 dollar to 12 dollar the quantity demanded of John falls. Then quantity demanded ~~for~~ decreases from 5 unit to 4 unit. Now, Adding the point A_1 and B_1 . we get the demand curve D_1D_1' .

Demand curve for Smith: when the price is 11 dollar the quantity demanded of John is 6 unit. If prices rises from 11 dollar to 12 dollar the quantity demanded of John falls from 6 unit to 5 unit. Adding the point A_2 and B_2 . Then we get the demand curve of Smith and that is D_2D_2' .

Demand curve for other Buyers: At the price of 11 dollar the quantity demanded of other buyer is 130 unit. when prices rises from 11 dollar to 12 dollar quantity demanded of other buyer decreases from 130 unit to 100 unit. Point A_3 indicates it. Now Adding the point A_3 and B_3 for getting the demand curve of other Buyer. D_3D_3' is the demand curve of other buyer.

Demand curve for all Buyers: At the price of 11 dollar the quantity demanded for all Buyer is 141 unit. when price rises from 11 dollar to 12 dollar quantity demanded of all buyer falls from 141 unit to 109 unit. Points A_4 indicates it. Now, Adding the point A_4 and B_4 for getting the demand curve of all buyer. D_4D_4' is the demand curve for all buyer.

Supply: Supply refers to the willingness and ability of sellers to produce and offer to sell different quantities of a good at different prices during a specific time period.

Quantity supplied: Quantity supplied is the number of units of goods that sellers are willing and able to produce and offer to sell goods at particular price during a time period.

Law of supply: The law of supply states that if the price of the good rises the quantity supplied of the good rises and as the price of the good falls, the quantity supplied of the good falls.

ceteris paribus

$P \uparrow \quad Q_s \uparrow$

$P \downarrow \quad Q_s \downarrow$

; P = Price

Q_s = Quantity supplied

Supply curve: The supply curve is a graph that illustrates how much of a good a seller would be willing to sell at different price other thing held constant.

A change in quantity supplied: A movement from one point to another point on the same supply curve caused by the change in the price of the good.

Supply schedule:

Price (€)	Q _s (units)
10	100
20	200
30	300
40	400

Supply curve:

Figure 1.

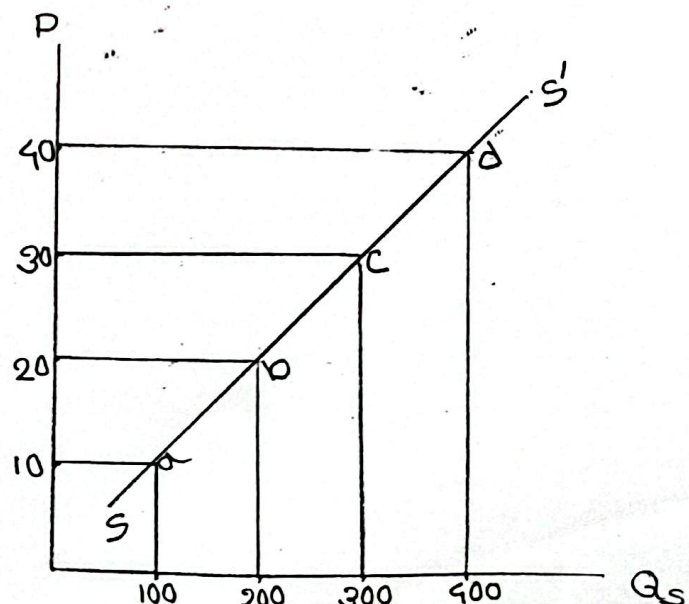


Figure 1, shows us a change in quantity supplied. The horizontal axis labeled 'quantity supplied' for a good and the vertical axis labeled price as dollars. when the price is 10 dollar the quantity supplied is 100 unit. Point a represent it. when the price increase from 10 to 20 dollar the quantity supplied is also increase from 100 unit to 200 unit. After that, when the price is 30 dollar the quantity supplied is 300 units. Point c indicate it. Now adding all the points (a, b, c, d) for getting the supply curve SS'. All points are follow the law of supply.

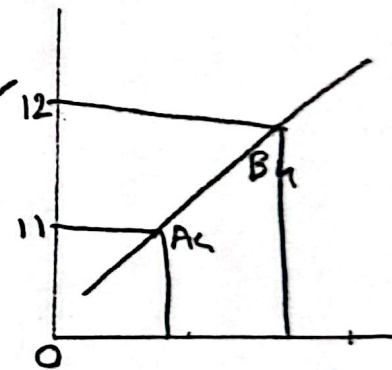
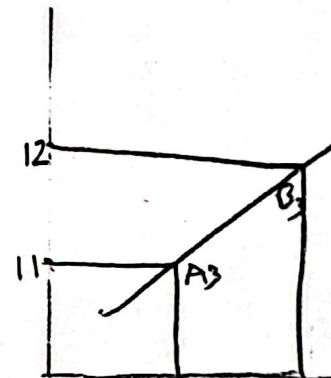
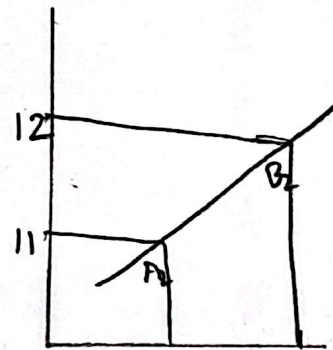
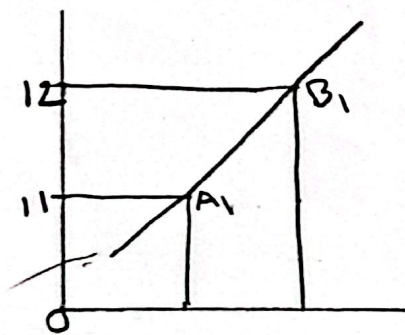


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Supply schedule: Supply schedule is a table showing how much of a product a seller would be willing to sell at different prices when other things held constant.

Price (P)	Quantity supplied (Q _s)
20	5
30	10
40	15
50	20

Factors cause the supply curve to shift.

- * Price of Relevant resources

- * Technology

- * Price of other Goods

- * Number of sellers

- * Expectations of Future price

- * Taxes and Subsidies

- * Government restriction

Price of relevant resources: Resources are needed to produce goods. For example, wood is needed to produce doors. If the price of wood falls producing doors become less costly. If the price of wood increase, producing doors becomes more costly.

Technology: An advanced in technology was said to refer to the ability to produce more output with the fixed amount of resources, reducing per unit production cost. If, the technological knowledge

Price of other goods: Think of a farmer who is producing wheat, suddenly the price of something he is not producing (say, corn) rises relative to wheat. The farmer might shift his farming away from wheat to corn. In other words, as the price of a corn rises relative to wheat, the farmer switches from wheat to corn production. We conclude that a change in the price of one good can lead to change in the supply of other goods.

Number of sellers: If more sellers begin producing a good, perhaps because of high profits, the supply curve will shift rightward. If some seller stops producing a good, perhaps because of losses, the supply curve will shift leftward.

Expectation of future price: If the price of a good is expected to be higher in the future, producers may hold back some of the product today. (If possible - perishables cannot be held back) Then they will have more to sell at the higher future price. Therefore the current supply curve will shift leftward.

Taxes and subsidies: Some taxes increase per unit costs. If the tax is increased the supply curve shifts leftward. If the tax is eliminated the supply curve shifts rightward.

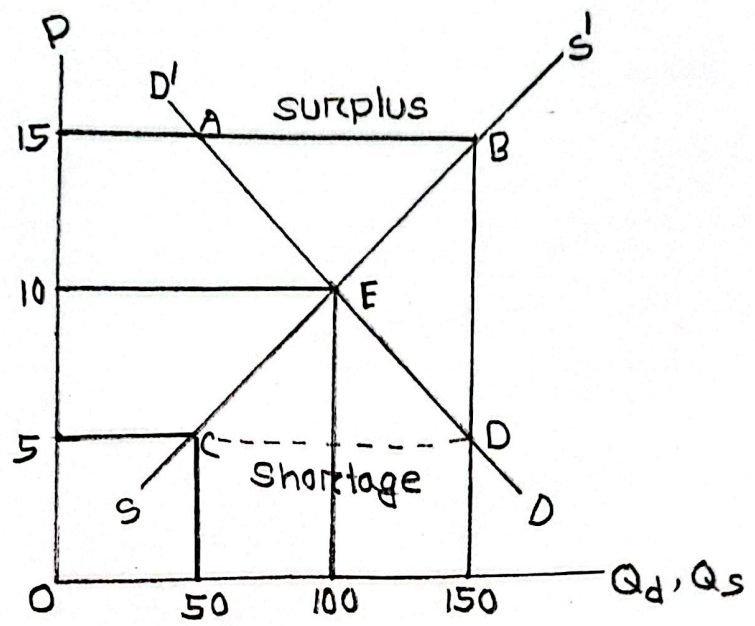
Subsidies have the opposite effect. If the government gives subsidies to the production process the supply will increase. Then the supply curve shifts rightward.

Government Restriction: Sometime government acts to reduce supply. Then the supply will decrease. Thus, the supply curve shifts leftward.

Market: An arrangement whereby buyers and sellers interact to determine the prices and quantities of a commodity. Some markets take place in physical locations, other markets are constructed over the telephone or are organized by computer and some markets now are organised on the internet.

Market equilibrium: A market equilibrium comes at the price at which quantity demanded equals quantity supplied. At the equilibrium there is no tendency for the price to rise or fall. The equilibrium price is also called market-clearing price.

Price (\$)	Quantity supplied (Q_s)	Quantity Demanded (Q_d)	Condition
15	150	50	surplus
10	100	100	equilibrium
5	50	150	Shortage



Surplus (Excess supply): A condition in which the quantity supplied is greater than the quantity demanded. Surplus occurs only at prices above equilibrium price. In figure 1, points AB represent surplus.

Shortage (Excess Demand): A condition in which the quantity demanded is greater than quantity supplied. Shortage occurs only at prices below equilibrium price. In figure 1, points CD represent shortage.

Equilibrium price (Market clearing price): The price at which the quantity demanded of the good equals the quantity supplied. In figure 1, equilibrium price is shown by point E.

Equilibrium quantity: The quantity that corresponds to equilibrium price. The quantity at which the amount of the good that buyers are willing and able to buy equals the amount that sellers are willing and able to sell, and both equal the amount actually bought and sold. In figure 1, equilibrium quantity is 100 units.

Disequilibrium price: A price other than equilibrium price. A price at which the quantity demanded does not equal the quantity supplied. In figure 1, the equilibrium price is 10 dollar and that's why the disequilibrium prices are 5 and 15 dollar.

Disequilibrium: A state of either surplus or shortage in a market. In figure 1, all the points are disequilibrium except point E.

Equilibrium: Equilibrium in a market is the price quantity combination from which buyers or seller do not tend to move away. In figure 1, equilibrium ~~price~~ is shown by point E.

*why does price fall when there is a surplus
In figure 1, there is a surplus at a price of \$15. The quantity supplied (150 units) is greater than the quantity demanded (50 units). Suppliers will not be able to sell all they had hoped to sell at \$15. As a result, their inventory will grow beyond the level they hold in

preparation for demand changes, sellers will want to reduce their inventories. Some will lower prices to do so, some will cut back on production, others will do a little of both. As shown in the exhibit, price and output tend to fall until equilibrium is achieved.

* why does price rise when there is a shortage? In Exhibit 1, there is a shortage at price of \$5. The quantity demanded (150 units) is greater than the quantity supplied (50 units). Buyers will not be able to buy all they had hoped to buy at \$5. Some buyers will bid up the price to get sellers to sell to them instead of to other buyers. Some sellers seeing buyers clamor for the goods, will realize that they can raise the price of the goods they have for sale. Thus, price and output tend to fall until equilibrium is achieved.